Guofeng Qian

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Google Scholar | LinkedIn

Canadian Permanent Resident; authorized to work in Canada.

Research profile

Structural engineer and researcher developing physics-informed digital twins for infrastructure condition assessment, deterioration prediction, and risk-informed decisions. Expertise in computational mechanics, scientific machine learning, structural monitoring, Bayesian inference, uncertainty quantification, and structural reliability. Experience spans government-supported research and monitoring of operating bridges.

Education

Ph.D. in Structural Engineering | University of California San Diego
January 2021 - December 2023 | Advisor: Prof. Michael D. Todd

Dissertation: Diagnostics and Prognostics of Pitting Corrosion in Large Civil Infrastructure Using Multi-Scale Simulation and Machine Learning.

M.S. in Structural Engineering | University of California San Diego
September 2019 - December 2020 | Health Monitoring and Nondestructive Evaluation

B.Eng. in Civil Engineering | Harbin Institute of Technology
September 2015 - June 2019

Exchange student | Monash University | January - June 2018

Engineering experience

Research and Development Engineer
LPI Ingenieurgesellschaft mbH, Hannover, Germany | September 2025 - August 2026

- Developed simulation- and monitoring-based bridge assessment workflows combining finite-element modeling, structural dynamics, uncertainty quantification, and long-term field measurements.
- Automated parametric SOFiSTiK simulations and output-only modal analysis with Python. Compared temperature-dependent dynamics of twin bridges, one with documented tendon breaks.
- Developed strain-based impact detection and combined strain, displacement, and inclination measurements. Interpretation supported a client decision to maintain a damaged steel bridge's closure.
- Developed acoustic-emission and distributed fiber-optic workflows for wire-break classification, crack localization, and crack-width estimation.

Research and teaching experience

University of California San Diego | La Jolla, United States

Postdoctoral researcher | January 2024 - August 2025

- Extended a USACE/ERDC-supported digital-twin program from individual structures to population-based diagnostics and changing environmental conditions.
- Developed domain-adaptation methods for damage diagnostics across structures and temperature conditions. On simulated Bonneville responses, gap MSE fell from 0.069 to 0.015 with target labels withheld during training.
- Developed normalizing-flow-based Bayesian updating for multiphysics corrosion models and integrated predictions with accelerated corrosion experiments.
- Presented semiannual sponsor reviews, developed proposal content, and mentored graduate and undergraduate researchers.

Graduate student researcher | January 2021 - December 2023

- Built a coupled mechano-electrochemical phase-field model of localized corrosion. Connected pit evolution with structural stress analysis and crack-initiation reliability of miter gates.
- Automated parametric simulations and developed autoencoder-based reduced models and probabilistic forecasts. Contributed simulation data and the baseline surrogate to physics-constrained morphology forecasting.
- Combined active learning, importance sampling, and machine-learning surrogates for gate-wide risk maps; the local reliability benchmark used 101 function evaluations versus 674 for AK-MCS.
- Developed two-stage approximate Bayesian computation and conditional invertible neural networks for updating corrosion models from inspection images and experimental observations.

Research assistant | January - December 2020

- Modeled composite wing-spar debonding in Abaqus, automated parametric simulations and result extraction, and developed a deep-learning surrogate for strain across damage locations and severities.

Teaching assistant | University of California San Diego

SE 168/268 Structural System Testing and Model Correlation | Spring 2022, 2023
Co-taught by Prof. Michael Todd and ATA Engineering.

- Led labs in test planning, hammer and shaker excitation, accelerometer instrumentation, synchronized acquisition, and data quality.
- Guided processing of frequency-response functions, coherence and reciprocity checks, modal identification, finite-element correlation, model reduction, and boundary-stiffness updating.

SE 130A/B Structural Analysis I and II | Summer 2021

Journal articles and book chapter

Guofeng Qian, Duncan Jay Fure, Long Wang, Zhen Hu, and Michael D. Todd. **Machine learning-enabled probabilistic hybrid model updating of pitting corrosion with application to hydraulic steel structures: an experimental validation**. Journal of Civil Structural Health Monitoring (Accepted). Open code.

Yichao Zeng, Zhao Zhao, **Guofeng Qian**, Michael D. Todd, and Zhen Hu. **Enhancing Bayesian Inference-Based Damage Diagnostics Through Domain Translation With Application to Miter Gates**. Journal of Mechanical Design, 147(6), 061701 (2025). doi:10.1115/1.4067719

Guofeng Qian, Jice Zeng, Zhen Hu, and Michael D. Todd. **Bayesian Model Updating of Multiscale Simulations Informing Corrosion Prognostics Using Conditional Invertible Neural Networks**. ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part B: Mechanical Engineering, 11(1), 011105 (2025). doi:10.1115/1.4065845

Guofeng Qian, Zihan Wu, Zhen Hu, and Michael D. Todd. **Pitting Corrosion Diagnostics and Prognostics for Miter Gates Using Multiscale Simulation and Image Inspection Data**. Structural Health Monitoring (2024). doi:10.1177/14759217241264291

Guofeng Qian, Zhen Hu, and Michael D. Todd. **Physics-Based Corrosion Reliability Analysis of Miter Gates Using Multi-Scale Simulations and Adaptive Surrogate Modeling**. Mechanical Systems and Signal Processing, 200, 110619 (2023). doi:10.1016/j.ymssp.2023.110619

David Najera-Flores, **Guofeng Qian**, Zhen Hu, and Michael D. Todd. **Corrosion morphology prediction of civil infrastructure using a physics-constrained machine learning method**. Mechanical Systems and Signal Processing, 200, 110515 (2023). doi:10.1016/j.ymssp.2023.110515

Ying Zhao, Mayank Chadha, Nicholas Olsen, Elissa Yeates, Josh Turner, Guga Gugaratshan, **Guofeng Qian**, Michael D. Todd, and Zhen Hu. **Machine learning-enabled calibration of river routing model parameters**. Journal of Hydroinformatics, 25(5), 1799-1821 (2023). doi:10.2166/hydro.2023.030

Guofeng Qian, Karnpiwat Tantratian, Lei Chen, Zhen Hu, and Michael D. Todd. **A probabilistic computational framework for the prediction of corrosion-induced cracking in large structures**. Scientific Reports, 12, 20898 (2022). doi:10.1038/s41598-022-25477-8

Book chapter

Guofeng Qian, Mayank Chadha, Zihan Wu, Zhen Hu, and Michael D. Todd. **The Use of Digital Twin Architectures in Structural Health Monitoring and Management of the Inland Waterways Navigation Civil Infrastructure**. In Condition Monitoring and Structural Health Monitoring of Infrastructure Systems, edited by Vistasp M. Karbhari and Farhad Ansari. Woodhead Publishing, pp. 675-718 (2026). doi:10.1016/B978-0-443-38315-1.00016-7

Conference contributions | 2026-2025

2026

Guofeng Qian, Jan-Hauke Bartels, Christoph Tomann, and Nico Dieckmann. **Case Study on Temperature Effects on Bridge Frequencies**. 12th European Workshop on Structural Health Monitoring (EWSHM), Toulouse, France (2026).

Jan-Hauke Bartels, **Guofeng Qian**, and Lasse Petersen. **Value of Information for Structural Health Monitoring of a Prestressed Concrete Bridge with Stress-Corrosion-Cracking Risk**. 13th International Conference on Structural Dynamics (EURODYN), Hannover, Germany (2026).

2025

Guofeng Qian, Haiwei Lin, Travis Fillmore, Hai Nguyen, Brian Eick, Zhen Hu, and Michael D. Todd. **Generalized Damage Diagnostics of Two Miter Gates Through Domain Adaptation**. 15th International Workshop on Structural Health Monitoring (IWSHM), Stanford, California (2025). doi:10.12783/shm2025/37569

Guofeng Qian, Duncan Jay Fure, Long Wang, Zhen Hu, and Michael D. Todd. **Pitting Corrosion Model Updating Using Experimental Data**. IWSHM, Stanford, California (2025).

Guofeng Qian, Jan-Hauke Bartels, and Michael D. Todd. **Unsupervised Damage Diagnostics with Data Normalization Using Experimental Data**. IWSHM, Stanford, California (2025).

Jan-Hauke Bartels, **Guofeng Qian**, and Michael D. Todd. **Matched Filtering-Based Sensor Fault Classification for Structural Health Monitoring Systems**. IWSHM, Stanford, California (2025).

Yichao Yang, Haiwei Lin, **Guofeng Qian**, Zhen Hu, and Michael D. Todd. **A Privacy-Preserving Framework Using Federated Learning for Structural Health Monitoring with Miter Gates Application**. IWSHM, Stanford, California (2025). doi:10.12783/shm2025/37568 **Best Paper (coauthor).**

Guofeng Qian, Jice Zeng, Zhen Hu, and Michael D. Todd. **Likelihood-Free Bayesian Inference for Localized Corrosion in Miter Gates**. Model Validation and Uncertainty Quantification, SEM Conference Proceedings, Springer (2025).

Guofeng Qian, Haiwei Lin, Zhen Hu, and Michael D. Todd. **Multi-Scale Multi-Mode Damage Simulation for Miter Gates**. Model Validation and Uncertainty Quantification, SEM Conference Proceedings, Springer (2025).

Conference contributions and research program

2024

Guofeng Qian, Zhen Hu, and Michael D. Todd. **Pitting Corrosion Monitoring Framework for Miter Gates with Digital Twin**. 11th European Workshop on Structural Health Monitoring (EWSHM), Potsdam, Germany (2024). doi:10.58286/29734

Guofeng Qian, Zihan Wu, Zhen Hu, and Michael D. Todd. **Multiscale Corrosion Damage Diagnostics and Prognostics for a Miter Gate**. Model Validation and Uncertainty Quantification, SEM Conference Proceedings, Springer (2024).

Yichao Zeng, Zhao Zhao, **Guofeng Qian**, Michael D. Todd, and Zhen Hu. **Damage Diagnostics of Miter Gates Using Domain Adaptation and Normalizing Flow-Based Likelihood-Free Inference**. Annual Conference of the PHM Society, 16(1) (2024). doi:10.36001/phmconf.2024.v16i1.4223

2023

Guofeng Qian, David Najera-Flores, Zhen Hu, and Michael D. Todd. **A hybrid surrogate modeling method for corrosion morphology prediction under non-stationary dynamic loading**. 14th International Workshop on Structural Health Monitoring (IWSHM), Stanford, California, pp. 643-650 (2023). doi:10.12783/shm2023/36797

Guofeng Qian, Zhen Hu, and Michael D. Todd. **Physics-Based Corrosion Reliability Analysis of Miter Gates Using Multiscale Simulations and Adaptive Surrogate Modeling**. Model Validation and Uncertainty Quantification, Volume 3, SEM Conference Proceedings, Springer, pp. 101-105 (2023). doi:10.1007/978-3-031-37003-8_16

Research proposal development | 2021-2025

USACE/ERDC, Innovative Technologies in Structural Health Monitoring for Condition Assessment and Reliability Prediction (W9132T-22-20014). Developed research concepts, proposed methods, preliminary results, and subsequent tasks for continuation proposals. Presented directions and progress at semiannual technical reviews.

- 2025: Multimode multiphysics damage prognostics using physics-based and data-driven simulation.
- 2025: Population-based damage diagnostics with multisource information transfer.
- 2024: Likelihood-free Bayesian model updating with synthetic image observations.
- 2023: Reliability analysis of localized corrosion using multiscale simulation.
- 2022: Multiphysics corrosion simulation and machine-learning surrogate modeling.

Mentoring, service, and skills

Research mentorship

Haiwei Lin, Ph.D. researcher, UC San Diego | Summer 2024 - Fall 2025

- Initiated a collaboration with Prof. Shabnam Semnani to connect the corrosion model with phase-field fracture. Mentored formulation, multiphysics implementation, interpretation, and research communication.

Vaibhav Pratap, M.S. independent study, UC San Diego | Spring 2023

- Provided technical guidance in reliability analysis and uncertainty quantification.

Edward Yu and Peter Le, NSF REU, UC San Diego | Summer 2022

- Guided cantilever-plate testing and MATLAB finite-element correlation, including accelerometer placement, modal properties, and rotational-spring updating to identify changes in connection stiffness.

Academic and professional service

Journal reviewer: ASCE-ASME Journal of Risk and Uncertainty, Part B; Journal of Computational Physics; Journal of Verification, Validation and Uncertainty Quantification; Journal of Structural Design and Construction Practice.

Society for Experimental Mechanics (SEM), 2022 - Present.
Earthquake Engineering Research Institute (EERI), 2022 - Present.

Selected honors and awards

- Best Paper, IWSHM 2025, federated-learning paper (coauthor).
- Distinguished Graduate Thesis, 2019.
- International Study Grant, Monash University, 2018.
- Best Paper, International Conference for Integration of Science, Technology and Society, 2016.
- "Xu Zengquan" Scholarship, 2016.

Technical skills

Machine learning: surrogate modeling, physics-informed learning, CNNs, autoencoders, Bayesian neural networks, Gaussian processes, normalizing flows, domain adaptation, adversarial learning.

Computational and probabilistic methods: finite-element and multiphysics simulation, Bayesian inference, model updating, Monte Carlo methods, uncertainty quantification, structural reliability.

Monitoring: modal analysis, system identification, signal processing, acoustic emission, distributed fiber-optic sensing.

Programming and software: Python, PyTorch, TensorFlow, MATLAB, Abaqus, COMSOL, FEniCSx, SOFiSTiK, Git, LaTeX.